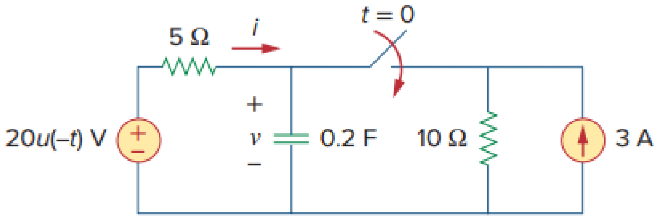
Chapter 7, Problem 11PP

Problem

The switch in Fig. 7.47 is closed at t = 0. Find $i(t)$ and $v(t)$ for all time. Note that $u(-t) = 1$ for $t<0$ and 0 for $t > 0$. Also, $u(-t) = 1 - u(t)$.

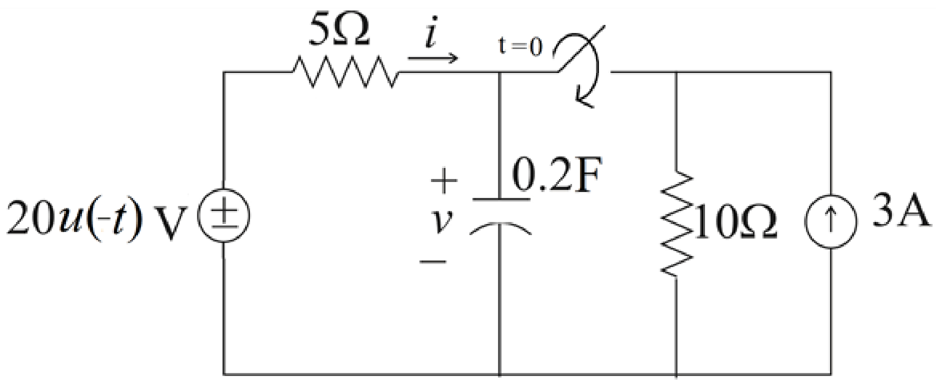
Fig. 7.47:



Step-by-step solution

Step 1 of 13

Given circuit diagram



Step 2 of 13

The resistor current i is discontinuous at $t=0$.

But capacitor voltage is continuous. Hence first find the capacitor voltage ' v'

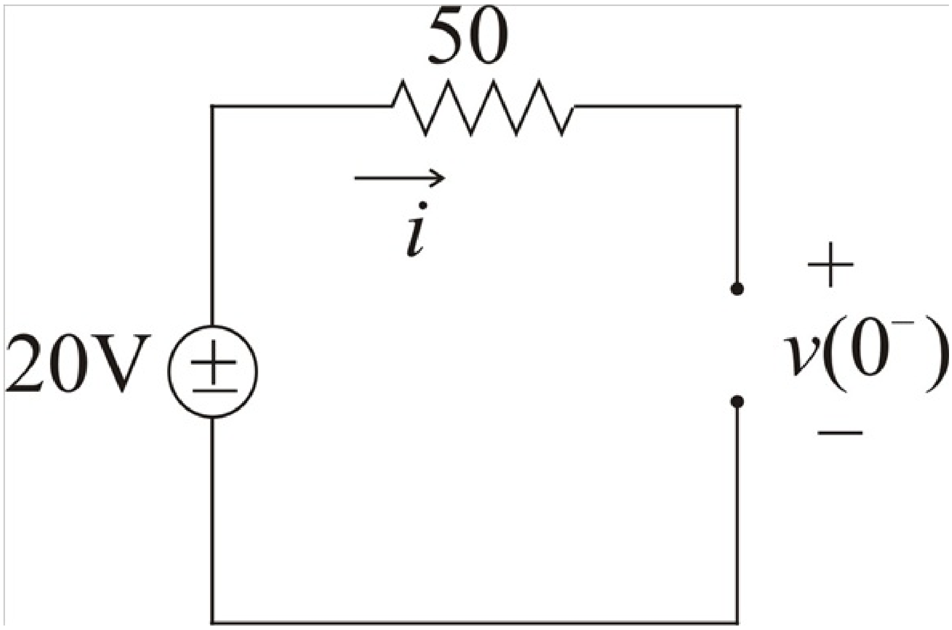
$$u(-t)=\begin{cases}1 & t<0\\ 0 & t>0\end{cases}$$

Switch is closed at $t=0$.

For $t<0$ capacitor acts like a open circuit for dc.

Hence equivalent circuit for $t<0$

Step 3 of 13



Step 4 of 13

From the above circuit $v_C(0^-) = 20\text{V}$

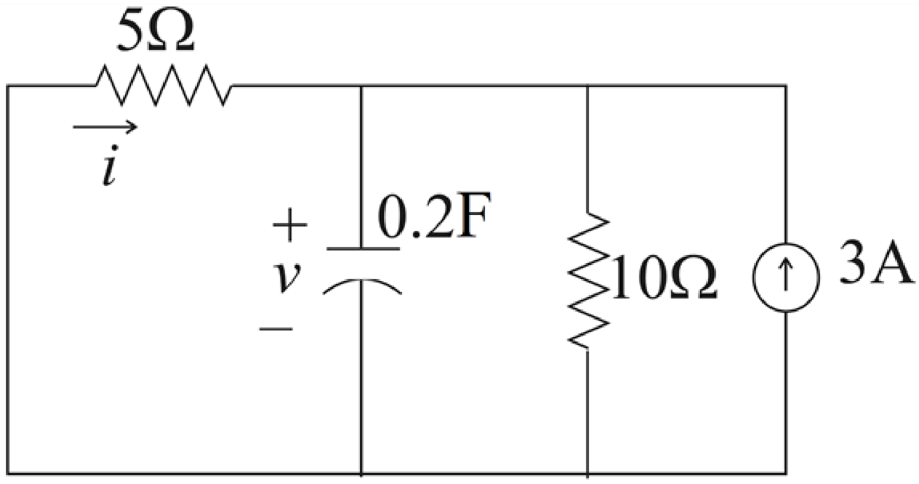
Capacitor voltage can't change instantaneously

$v(0) = v(0^-) = v(0^+) = 20\text{V}$

For $t < 0$, $i = 0$

Step 5 of 13

For $t > 0$ switch is closed, equivalent circuit for the given circuit is as below

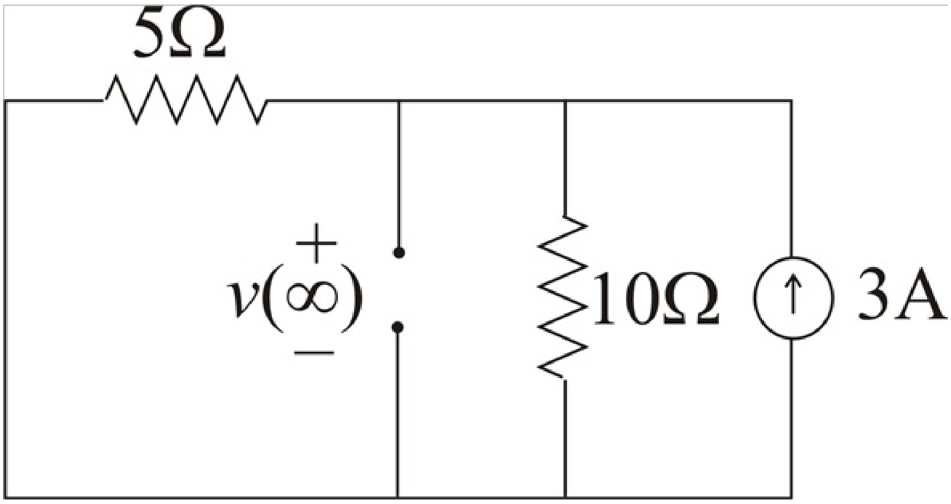


Step 6 of 13

After a long time, the circuit reaches steady state and the capacitor acts like an open circuit again.

Hence the circuit becomes as below

Step 7 of 13



Step 8 of 13

By using current division rule we can write

$$I_{10\Omega} = 3 \times \frac{5}{5+10}$$

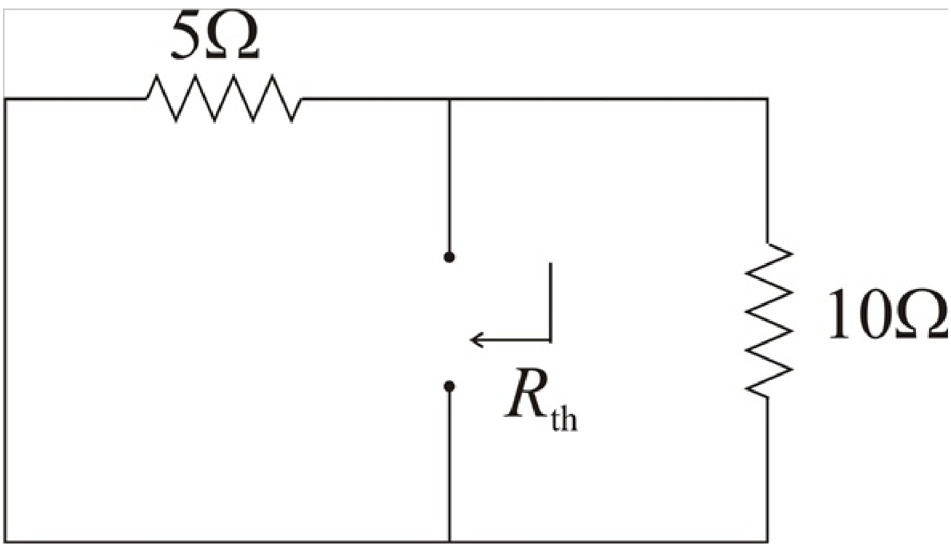
$$I_{10\Omega} = 1\,\text{A}$$

$$\therefore \; v(\infty) = 10\left(I_{10\Omega}\right)$$

$$v(\infty) = 10\,\text{V}$$

Equivalent or Thevenin's resistance can be found by open circuiting current source and short circuiting voltage source

Step 9 of 13



Step 10 of 13

$$R_{th} = 5 \,\|\, 10$$

$$R_{th} = \frac{5 \times 10}{10+5}$$

$$R_{th} = \frac{10}{3}\,\Omega$$

Step 11 of 13

Time constant $\tau = R_{th}C$

$$\tau = \frac{10}{3}\times 0.2$$

$$\tau = \frac{2}{3}\text{sec}$$

Thus capacitor voltage is

$$v(t) = v(\infty) + \left[v(0) - v(\infty)\right]e^{-\frac{t}{\tau}}$$

$$v(t) = 10 + (20-10)e^{-\frac{t}{\frac{2}{3}}}\text{V} \quad \text{for} \quad t>0$$

$$v(t) = 10 + 10e^{-\left(\frac{3}{2}\right)t}\text{V} \quad \text{for} \quad t>0$$

$$v(t) = 10\left(1+e^{-1.5t}\right)\text{V} \quad t>0$$

Step 12 of 13

For $t>0$ applying KVL to left loop we can write

$$5i + v(t) = 0$$

$$i = -\frac{v(t)}{5}$$

$$i = \frac{-10\left(1+e^{-1.5t}\right)}{5}$$

$$i = -2\left(1+e^{-1.5t}\right)\text{A} \quad t>0$$

Step 13 of 13

Therefore $i(t)$ and $v(t)$ for all time

$$\boxed{i(t) = \begin{cases} 0\text{A} & t<0 \\ -2\left(1+e^{-1.5t}\right)\text{A} & t>0 \end{cases}}$$

$$\boxed{v = \begin{cases} 20\text{V} & t<0 \\ 10\left(1+e^{-1.5t}\right)\text{V} & t>0 \end{cases}}$$